Bouncing Ball Steps

Time limit: 1.00 s **Memory limit:** 512 MB

There is a ball at the top-left corner of an $n \times m$ grid. The rows of the grid are numbered $1, 2, \ldots, n$, and the columns are numbered $1, 2, \ldots, m$.

The ball is initially moving diagonally away from the top-left corner. At every step, it moves one cell. Whenever the ball hits the border of the grid, it changes its direction.

What is the location of the ball after k steps and how many times has it changed direction?

Input

The first line has an integer t: the number of tests.

After this, there are t lines. Each line has three integers n, m and k: the size of the grid and the number of steps.

Output

For each test, print three integers: the location of the ball and the number of direction changes.

Constraints

- $1 \leq t \leq 1000$
- $2 \leq n, m \leq 10^9$
- $0 \leq k \leq 10^{18}$

Example

Input

```
6
3 4 0
3 4 1
3 4 2
3 4 3
3 4 4
42 1337 123456789
```

Output

```
1 1 0
2 2 0
3 3 1
2 4 2
1 3 3
34 300 3101295
```